

Sulaiman Dawood, M.S.

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CURRENT APPOINTMENT Senior Project Engineer
American Contracting and Environmental Services Inc. Maryland, USA

RESEARCH MISSION To advance sustainable, high-performance concrete systems through fundamental and applied research in cementitious materials, enabling low-carbon, resilient, and next-generation infrastructure on Earth and in extraterrestrial environments.

RESEARCH INTERESTS ♦ Advanced Concrete Materials ♦ Low-Carbon and Sustainable Cementitious Systems ♦ Geopolymer and Alkali-Activated Materials ♦ 3D Printed and Additively Manufactured Concrete ♦ Nanoindentation and Micromechanical Characterization ♦ Lunar and Extreme-Environment Concrete ♦ Durability and Performance-Based Concrete Design ♦ Sustainable Construction Materials ♦ Microstructure–Property Relationships

Skills

- Statistical analysis
- MATLAB
- Scanning Electron Microscope (SEM)
- Gaussian Mixture Model (GMM)
- Nano-Indentation
- Energy Dispersive X-ray Spectroscopy. (EDS)
- Python
- Data Analytics
- X-ray Diffraction (XRD)

Accomplishments

♦ **Thesis**

1. Dawood S.,"Building Beyond Earth: A Micromechanical and Microstructural Study of Lunar Concrete," ***M.S. Thesis***, Clarkson University, Potsdam, NY, 2024.
    - Objectives: To investigate the micromechanical behavior and microstructural evolution of lunar-regolith-based concrete systems to enable structurally reliable and resource-efficient extraterrestrial construction.
    - Methodology / Summary: Experimental testing combined with microstructural characterization (SEM, XRD) and micromechanical modeling was employed to evaluate strength development, bonding mechanisms, and failure behavior of lunar concrete under simulated extraterrestrial conditions.
2. Dawood S.,"Energy Efficiency Design Concept of Modern Houses and building'' ***B.Sc. Final Year Project Thesis***, University of Engineering and Technology Pakistan, Taxila, 2021.
    - Objectives: To develop energy-efficient residential building design strategies that reduce operational energy consumption while maintaining occupant comfort in modern housing.
    - Methodology / Summary: The study integrated architectural design optimization, material selection, and passive energy-efficiency techniques, supported by comparative energy performance analysis of conventional and optimized building models.

♦ **Journal Articles**

- Dawood, M.S., Collins, P.J., Radlińska, A., and Thomas, R.J. "Microstructure and Micromechanical Properties of Lunar Regolith Simulant-Based Geopolymers." Journal of Aerospace Engineering (ASCE), 2026; published online November 17, 2025. (https://doi.org/10.1061/JAEEEZ.ASENG-6302)
- Dawood, M.S., and Thomas, R.J. "On the Evolution of Microstructure and Micromechanical Properties in Belitic Calcium Sulfoaluminate Cement." Preprint available on SSRN; Under review.
- Dawood, M.S., Collins, P.J., Radlińska, A., and Thomas, R.J. "Micromechanical Properties of Lunar Regolith Simulants and Their Constitutive Minerals." In preparation.
- Dawood, M.S., Collins, P.J., Radlińska, A., and Thomas, R.J. "Nanoindentation Investigation of Geopolymer Lunar Concrete from Exposure in Low Earth Orbit." In preparation.

♦ **Conference Paper**

- Ahmad, W., Dawood, M.S., Collins, P.J., Radlińska, A., and Thomas, R.J. "Effects of Concrete Exposure in Low Earth Orbit." Earth and Space 2026, conference proceedings. Accepted for publication, 2026.
- Dawood, M.S., Collins, P.J., Radlińska, A., and Thomas, R.J. "Micromechanical and Microstructural Analysis of Lunar Concrete." Earth and Space 2024: Engineering for Extreme Environments, conference proceedings, 2024. (https://doi.org/10.1061/9780784485736.079)

♦ **Oral presentations**

- Ahmad, W., Dawood, M.S., Collins, P.J., Radlińska, A., and Thomas, R.J. "Effects of Concrete Exposure in Low Earth Orbit," ASCE Earth and Space 2026 Conference, College Station, TX, USA, April 2026.
- Dawood M.S., Collins, P.J., Radlińska A. and Thomas, R.J. "Micromechanical and microstructural Analysis of lunar concrete," Spring 2024 ASCE Space Conference, Miami, FL, USA, April 15-19, 2024.
- Dawood, M.S., and Thomas, R.J. "Micromechanical And Microstructural Analysis of Lunar Concrete," ACI 123 Research in Progress Session, Fall 2023 ACI Convention, Boston, MA, USA, October 29-November 2, 2023.(https://www.youtube.com/watch?v=Kqzoa17t3yI)

♦ **Poster presentations**

- Dawood, M.S., and Thomas, R.J. "Micromechanical Properties of Lunar Concrete," Center for Advanced Materials Processing (CAMP) Technical Meeting, Clarkson University, Corning, NY, USA, May 31-June 1, 2023 .
- Dawood, M.S., and Thomas, R.J. "Micromechanical Properties of Lunar Concrete," Research and Projects Showcase (RAPS), Clarkson University, Potsdam, NY, USA, April 14, 2023.

♦ **Accepted Abstracts**

- "Micromechanical Characterization of Lunar Regolith Simulants for Future Lunar Infrastructure." ASCE 2027, Abstract No. 2114. Accepted May 2026.
- "Micromechanical Assessment of Lunar Regolith Simulants for ISRU-Based Lunar Construction." Lunar Surface Innovation Consortium Spring Meeting. Accepted March 2026.

Education

May 2024 **M.S., Civil and Environmental Engineering, Clarkson University (Potsdam, NY)**
Thesis: "Building Beyond Earth: A Micromechanical and Microstructural Study of Lunar Concrete"
Advisor: Robert J. Thomas
Graduated with 4.0/4.0 GPA.
Graduated in Top 1% of Class

May 2021 **B.Sc., Civil and Environmental Engineering, University of Engineering and Technology (UET Taxila)**
Graduated with 3.62/4 GPA.
Graduated in Top 1% of Class

Scholarships and Awards

- American Concrete Institute Emerging Leaders Alliance, selected ACI representative, 2026 Emerging Leaders Conference, Pittsburgh, PA; sponsored registration valued at 1,500 USD
- NASA Funded Scholarship for whole MSc tenure; 150194 USD
- Clarkson Diversity, Equity & Inclusion Distinguished Scholar.
- ACI Travel Stipend for ACI Fall Conference Boston; 1000 USD
- Clarkson University Graduate Student Association(CUGSA); Travel Grant for Conference.
- Teacher Foundation Scholarship, 4 years of Bachelor; 10000 PKR

Experience

2026 – Present **Senior Project Engineer**
2024 – 2026 **Project Engineer American Contracting And Environmental Services Inc. (ACE), Maryland**

- Lead engineering and construction coordination across $22M multi-phase industrial projects, working with architects, engineers, subcontractors, vendors, and owners to resolve field, design, and execution issues.
- Oversee construction administration including RFIs, submittals, shop drawing review, change orders, project documentation, and compliance with specifications and project requirements.
- Lead startup and commissioning support for heavy equipment systems, including coordination of I/O signal mapping, signal validation, and verification of integration between field instrumentation and SCADA systems.
- Lead the organization and technical review of operational datasets used for troubleshooting, performance verification, commissioning analysis, and reliable system startup.
- Conduct field inspections, document deficiencies, prepare site observation reports, and support timely corrective action across civil, structural, envelope, and mechanical scopes.
- Develop project dashboards tracking safety, quality, cost, and schedule KPIs; contributed to securing over $1M in approved change orders and improving project cost alignment.
- Support preconstruction planning, scope development, bid documentation, and design clarifications to strengthen constructability, documentation, and implementation readiness.

2024 – Present **Research Collaborator, Thomas Research Group, Department of Civil and Environmental Engineering, Clarkson University (independent of employment)**

- Analysis, interpretation, and visualization of experimental data for ongoing studies of lunar regolith-based binders and belitic calcium sulfoaluminate cement
- Drafting and editing of technical reports and journal manuscripts, including the MISSE-15 low-Earth-orbit exposure study
- Training junior research personnel in micromechanical characterization and data processing

2023 – 2024 **Instructor, Department of Civil and Environmental Engineering Clarkson University**

- Led instruction for 75 students in ES100 (Introduction to the Use of the Computer) at Clarkson University, emphasizing MATLAB & Simulink proficiency.

- Dedicated 20 weekly hours, ensuring a high-quality learning experience and fostering student engagement.

2022 – 2024 **Research Assistant, Department of Civil and Environmental Engineering Clarkson University**

- Demonstrated expertise in lunar regolith simulant-based geopolymer concrete, achieving compressive strengths approaching 40 MPa and generating data relevant to structurally viable lunar construction materials.
- Explored key mixture-design and curing variables influencing geopolymer lunar concrete, generating data relevant to future lunar construction, and contributed to ongoing studies of cementitious and regolith-based materials following Low Earth Orbit exposure on the International Space Station (ISS) as part of MISSE-15.
- Conducted research on material development, curing conditions, and mixture proportioning for geopolymer systems with additive manufacturing applications in mind.
- Utilized advanced characterization techniques including SEM-EDS, nanoindentation, and optical microscopy to evaluate microstructure and micromechanical behavior.

2021 – 2022 **Assistant Project Manager, Marco Polo Builders Pvt. Limited**

- Spearheaded the management of roadworks projects valued at 24 million while using project management tools such as P6 Primavera and Blue Beam that resulted in a 15% increase in project efficiency and a notable 10% reduction in material wastage.
- Collaborated with diverse team of clients, subcontractors and other stakeholders through meetings and tracked daily project related activities.

2020 – 2020 **Project Development Intern, World Youth Alliance**

- Drafted compelling proposals, facilitating successful project acquisition by effectively communicating project objectives.
- Spearheaded lead generation initiatives, resulting in a substantial increase in project opportunities and successful lead conversions on different projects completed during my tenure.
- Orchestrated two successful corporate networking events of 900+ individuals fostering key connections with multiple global companies, donors and alumni for sponsor engagement.

2020 – 2020 **Civil Engineering Intern, Education Engineering Cell, Department of Education**

- Coordinated project activities, ensuring effective communication and documentation for inspections on sites.
- Acted as a liaison between project stakeholders and provided necessary documentation for inspections.
- Utilized advanced Revit & AutoCAD skills to optimize building layouts, enhancing project efficiency and decreasing the project delivery rate by 6 months.

2020 – 2020 **Project Engineering Intern, Bunyad Developers & Construction Services**

- Coordinated project activities, ensuring effective communication and documentation for inspections on sites. Implemented Earned Value Cost Management for accurate monthly project estimates, enhancing financial forecasting accuracy.
- Analyzed survey results for subcontractors on 2 projects using data analytics, contributing to informed decision-making.
- Streamlined project initiation by actively contributing to the tendering process, including project quotations, schedules, and legal documentation.

Organizational Experience

2020 – 2020 **Amal Career Prep Fellow, Amal Academy (co-sponsored by PepsiCo and Stanford University)**

- Graduated with Honors after submitting a final project proposal in Engineering Entrepreneurial course.
- Developed business skills that included communication, leadership, problem solving and team work from instructors of Stanford Graduate School of business.

2020 – 2021 **Software Consultant UET Taxila Chapter, Institute of Civil Engineers (ICE)**

- Arranged multiple events in which around 500 students were taught AutoCAD, Autodesk Revit, ArcGIS and SAP 2000.
- Responsible for arranging keynote speakers, event management, operations, vendor management and logistics for more than 7 events.

2020 – 2020 **Leadership as Practice, Global Citizen Year Academy(GCYA), CA, USA**

- Graduated intensive 12-week leadership experience that equips students worldwide with powerful skills for a lifetime of social impact.
- Been part of a global cohort of over 250 students coming from 80+ countries.
- Studied the practice of leadership while doing research about my local community and crafted my own social impact proposal meant to develop my community.

Organizational Memberships

- American Concrete Institute (ACI)-(Member Committees)
    - 123-00 Research and Current Developments
    - 565-00 Lunar Concrete
    - 88-S805 Student Leadership Council
- Member American Society of Civil Engineers (ASCE)-(Member Committee)
    - Space Engineering and Construction (SEC) Technical Committee of the ASCE Aerospace Division (ASCE) – Construction & Materials subgroup
- Student Member International Association of Engineers (IAENG).
- Certified Member World Youth Alliance (WYA).
- Ex-Student Member Institute of Civil Engineers(ICE), UK
- Ex-Student member of Construction Management Association of America (CMAA).

Professional Service

- 25 verified peer review records across 14 manuscripts for five journals (Web of Science)
- Peer reviewer, Advances in Space Research; official Journal of the Committee on Space Research (COSPAR)
- Peer reviewer, American Society of Civil Engineers (ASCE) Journal of Aerospace Engineering
- Peer reviewer, Acta Astronautica, official journal of the International Academy of Astronautics
- Peer reviewer, American Concrete Institute (ACI) Structural Journal
- Peer reviewer, American Concrete Institute (ACI) Materials Journal

MOOC's & Certifications

- Project Management Professional (PMP), Project Management Institute (PMI), 2026
- OSHA 30-Hour Construction Training Course, 2024

- Python Environment and Syntax, Credly-Clarkson University , 2023
- Construction Project Management, Coursera-Columbia University, 2021
- Initiating and Planning Projects, Coursera-University of California Irvine, 2020
- Budgeting and Scheduling Projects, Coursera-University of California Irvine, 2020
- Six Sigma Principles, Coursera-Kennesaw State University, 2022
- Understanding Research Methods, Coursera-University of London, 2021
- Leadership as Practice, Global Citizen Year Certification, 2021
- Presenting your work with Impact, Future Learn-University of Leeds and Institute of Coding, 2020
- Manage a Project with Digital Tools, Google Applied Digital Skills, 2020

References

- Prof. Robert J. Thomas (Advisor): rthomas@clarkson.edu
- Prof. Behzad Behnia: behnia_b@utpb.edu
- Prof. Aleksandra Radlińska: azr172@psu.edu
- Mr. Michael R. Fiske: Michael.R.Fiske@nasa.gov
- Prof. Holly Shulman: Holly.Shulman@BlueStar-AM.com

Other Technical Skills

♦ AutoCAD ♦ Revit ♦ Civil 3D ♦ Navisworks ♦ Bluebeam ♦ Plan Swift ♦ P6 Primavera ♦ MS Excel ♦ MS Word ♦ MS PowerPoint ♦ Procore ♦ ETABS